Computer Science & IT



C Programming

Control Flow Statement

Lecture No. 05



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Recap of Previous Lecture



- Topic — loop ✓
- Topic
- Topic
- Topic
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Nested for loop

Mathematcal Aspect of for loop

Slide

#Q. consider the folllowing program

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k=k-(--j);
    for (i=0; i<5; i++)     {
        switch(i+k)         {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

The Number of times printf executed is____

$2*3/4+2.0/5+8/5$

$6/4+2.0/5+8/5$

$1+0.4+1=2.4$

$j=2.4$ = integer

$j=2$ ✓

predecrement

$k=k-(--j)$

$=0-(1)$

$k=-1$

#Q. consider the following program

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k=k-(--j);
    for (i=0; i<5; i++)     {
        switch(i+k)         {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

The Number of times printf executed is ___

		i+k	No. of printf
i = 0	k = -1	-1	1
i = 1	k = -1	0	1
i = 2	k = -1	1	3
i = 3	k = -1	2	3 ✓
i = 4	k = -1	3	2
			total = 10

No. of times printf executed is 10;



for loop

UB-LB+1 ← Not applicable

valid

```
int i;
for(i=1; i<=10; i=i+2){
    printf("I am a good student");
}
```

Iteration 1
1<=10

Iteration 2
3<=10

Iteration 3
5<=10

Iteration No 4
7<=10

Iteration 5
9<=10

Iteration 6
11<=10
false

Slide



for loop

i	i<=11	i=i+2
1	1<=11	✓
2	3<=11	✓
3	5<=11	✓
4	7<=11	✓
5	9<=11	✓
6	11<=11	✓

int i

```
for (i=1; i<=11; i=i+2) {
    printf("I am good student");
}
```

Slide



for loop

variable

addihve

```
int i;
for (i=1; i<=n; i=i+2){
    printf("I am a good student");
}
```

$\lceil 5.1 \rceil = 6$

$\lceil 5.9 \rceil = 6$

n → Even: $\frac{n}{2}$

$\frac{10}{2} = 5$

n odd $\left\lceil \frac{n}{2} \right\rceil$

$\left\lceil \frac{11}{2} \right\rceil = \lceil 5.5 \rceil = 6$

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for loop

uppar wala integer

Celing value : $\lceil 5.1 \rceil = 6$, $\lceil 5.2 \rceil = 6$

floor value $\lfloor 5.1 \rfloor = 5$

$\lfloor 5.9 \rfloor = 5$

Neeche wala integer



for loop

$\lfloor \log_2 24 \rfloor$

$= \lfloor 4._ \rfloor$

$= 4$ ✓

```
int i
for( i=2; i<=24; i=i*2){
    printf("I am good student");
}
```

	i=2	i=i*2
1	2<=24 ✓	4 ✓
2	4<=24	8 ✓
3	8<=24	16 ✓
4	16<=24	32 ✓
5	32<=24	∝

false

terminate

for Loop will run 4 times

Slide



for loop

```
for (i=2; i<=129; i=i*2){
    printf ("I am a good student");
}
```

Slide



for loop

i = i * 2

i = 2 ; $2^1 <= 129$; $i = 2^2$ ✓

$2^2 <= 129$, $i = 2^3$ ✓

$2^3 <= 129$; $i = 2^4$ ✓

$2^4 <= 129$, $i = 2^5$ ✓

$2^5 <= 129$, $i = 2^6$ ✓

$(64)2^6 <= 129$, $i = 2^7$

$128\ (2^7 < 129)\ 2^8$ ✓

$2^8 <= 129$ ✗

256 false

7 times Condition is true

Slide

k^{th} iteration $2^k <= n$

$\log_2$ both side

$\log_2 2^k <= \log_2 n$

$k \log_2 2 <= \log_2 n$

$k <= \log_2 n$



for loop

```
for (i=2; i<=n; i=i*2) {
    printf ("I am a good student")
}
```

$$\lfloor \log_2 n \rfloor = \lfloor \log_2 129 \rfloor$$

Observation



for loop

H.W

formula ??

```
int i
for (i=1; i<=n; i=i*2) {
    printf("I am a good student");
}
```

different values of n and then find formula by observation.

Slide



for loop

```
#include<stdio.h>
int main() {
    int i;
    int Sum = 0;
    for (i=1; i<=10; i++) {
        Sum = Sum + i;
    }
    printf("%d", Sum);
}
```

output _____

$Sum = 1 + 2 + 3 + 4 \ldots + 10$

$= \frac{n(n+1)}{2} = \frac{10 \times 11}{2}$

$n = 10 \quad = 55$

$0 + 1 - Sum = 1$

$Sum = 1 + 2$

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for loop

$$1+2+3+4\ldots+n=\frac{n(n+1)}{2}$$

Sum of n natural No.

Slide



for loop

* Hold, Complete Karenge

2nd problem

Nested loop का

if else Nested

for loop can be nested

function can also nested

Nested if

if statement give

inside of if statement

another if.

Slide



for loop

Tomorrow | Nested for loop : | loop inside loop

- Nested loop - for (i = 1 ; i <= 2 ; i++) {

Nesting of loop

for (j = 1 ; j <= 2 ; j++) {

printf ("%d" , i+j) ;

} }

Slide



for loop

+ve Remainder

2 / -8

4 + 2 % -8

+ve

4 + 2 = 6

-2 / -8

Negative

4 + (-2) % -8

4 + (-2) = 2

Slide



for loop

$i=1$, $j=j*j=1*1$. $Sum=0+1^2$

$i=2$ $j=2$, $j=2*2=2^2$ $Sum=0+1^2+2^2$

$i=3$ $j=3$ $j=3*3=3^2$ $=Sum=0+1^2+2^2+3^2$

$i=10$

$Sum=1^2+2^2+3^2+\ldots+10^2$ Sum of square

$$= \frac{n(n+1)(2n+1)}{6} = \frac{10\times11\times21}{6}$$

$$=385$$

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Sum of n natural No. square

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for loop

wrong

```
#include<stdio.h>
int main(){
int i, j=1, Sum=0;
for(i=1; i<=10, i++){
j=j*j; //j: 1^2, 4^2.
Sum=Sum+j; Sum=1^2+2^2+
j=j+1;
}
printf("%d", Sum);
}
```

wrong

two times modify

Output:

(A) 385

(B) 55

(C) 1 to 10

(D) 10,000

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```
# include<stdio.h>
int main(){
    int i, j=1, Sum=0;
    for(i=1; i<=10; i++){
        Sum=Sum+(j*j);
        j=j+1;
    }
    printf("%d", Sum);
}
```

$1^2+2^2+3^2+4^2+\cdots+10^2$

$$\frac{n(n+1)(2n+1)}{6}$$

$$= 385$$



Question

#Q. The number of times loop executed is

```
#include<stdio.h>
int main(){
int i,count=0;
for (i=1; i<=100; ){
     printf("\nI am a good Student");
    i = i+3;

}
return 0 ;
}
```

A. 33
B. 34
C. 100
D. 10

Slide



Question

#Q. The number of times loop executed is

```
#include<stdio.h>
int main(){
int i,count=0;
for (i=1; i<=100; i=i*2 ){
     printf("\nI am a good Student");

}
return 0 ;
}
```

A. 6
B. 7
C. 8
D. 10

The number of times value will be printed?

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Question

#Q.

What is the output of the program

```
#include<stdio.h>
int main(){
     int i,j,sum=0;
      for (i=1; i<=10;i++){
            j = j*j;
            sum= sum+j;
      }
      printf("%d",sum);
      return 0 ;
}
```

A. 385
B. 55
C. 10100
D. 10000

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2 mins Summary

Topic

Topic

Topic

Topic

Topic

Loop can be increment by ~~i=2~~ i = i + 2

Incremented by i = i * 2.

Sum of n natural No. $\frac{n(n+1)}{2}$

Sum of square n No. $\frac{n(n+1)(2n+1)}{6}$

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THANK - YOU